JAMES GRAY

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EDUCATION

University of Warwick | BEng Computer Systems Engineering (Year in Industry at QRT) Sept. 2022 – June 2026

- **Grade: First (81.1%)**
- **Award:** Exceptional Achievement in Second Year
- **Key Concepts:** AI, ML, Operating Systems, Networks, Computer Architecture, FPGAs & SoCs, Compilers, Databases, Signal Processing, Analogue Electronic Design, Maths

Reading School Sept. 2019 – July 2022

- **Grades: A*A*A*** (Maths, Physics, Computer Science)
- **Award:** Computer Science Award

SKILLS

- **Languages:** C, C++, C#, Python, Java, Haskell, MATLAB, TypeScript, JavaScript, SQL, HTML, CSS, Verilog
- **Technologies:** React, PyTorch, HLS, Docker, Linux

WORK EXPERIENCE

Qube Research & Technologies (QRT) | Quantitative Technology Intern June 2024 – June 2025

- Designed and built a **C++ process-management service** that lets support teams control and monitor system processes via a UI and code, **eliminating manual SSH interaction**.
- Introduced an **infrastructure-as-code** initiative that auto-generates environment-specific configurations and wires them into the process-management service, establishing a **single source of truth**.
- Reworked the **CI/CD pipeline** to support C# builds and tests on Linux, raising code coverage and **unblocking Linux developers**.
- Built an **LLM support assistant** using **RAG** over company-specific knowledge in a **reasoning and act loop** for multi-step reasoning, context access, and output verification, improving answer accuracy and quality.

MyTutor | A-Level and GCSE Tutor Jan. 2023 – Present

- **Communicated complex concepts in simple terms** to help A-Level and GCSE students master subject material.

KEY PROJECTS

Deep Learning for Efficient Sleep Staging with PPG Data (Python, PyTorch, Deep Learning, Time-Series)

- Pinpointed the **architectural bottlenecks** driving inference cost in **state-of-the-art** PPG AI sleep-staging models.
- Designed and implemented **Mamba-based** and **efficient-attention-based** deep learning architectures in PyTorch, which held accuracy (κ **= 0.74**) across **3 clinical datasets**.
- Raised inference throughput **3.8x** while cutting **VRAM 53%** and **model size 57%**, making deployment far more scalable.

Conjugate Gradient Optimisation on a 3D Mesh (C, AVX-256, OpenMP, Cache Optimisation)

- Optimised a 3D-mesh conjugate gradient solver with **AVX-256**, OpenMP, and locality refactoring for an **8.88x speedup**.

Mini C Compiler (C++, LLVM, Compilers)

- Built a **top-down recursive-descent** lexer and parser in C++, generating code via **LLVM IR**, for a subset of C.
- Produced colourised, informative error diagnostics **more detailed than mainstream C compilers**.

Multithreaded Packet Sniffer (C, Networking, Multithreading)

- Implemented a thread-safe packet queue with pthreads feeding a thread pool to detect domain blacklist violations, **SYN floods**, and **ARP cache poisoning** across **1,000,000s of packets without loss**.
- Validated the program was **free of memory leaks and race conditions** with **Valgrind, Helgrind, and GDB**.

FPGA Pacman (Verilog, Vivado, Nexys 4 FPGA)

- Implemented Pacman's game logic (movement, collisions, and scoring) as **hardware state machines**, taking the **top score in the year**.
- Rendered the game from sprites in on-chip **BRAM**, with custom frame-drawing logic and a **real-time VGA driver built in hardware**.

Other Projects

- Resistor Image Scanner (Python, OpenCV, K-means, Flask), Magnetic Electron Trap simulation (Python, SciPy, **20x speedup**), Movie Information Viewer (Java, from-scratch data structures), Gig Management System (Java, PostgreSQL), Discord Drive cloud storage (Python, Quart), Rhythm-game Discord bot, Maze & Circuit Solvers, Line-Following Buggy (C), Self-Balancing Robot (Simulink).